International Journal of Forecasting 41 (2025) 111–127

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

Local and global trend Bayesian exponential smoothing models

Slawek Smyl [a], Christoph Bergmeir [b,c,*], Alexander Dokumentov [d], Xueying Long [c], Erwin Wibowo [c], Daniel Schmidt [c]

[a] Walmart Labs, United States of America
[b] Department of Computer Science and Artificial Intelligence, University of Granada, Spain
[c] Department of Data Science and Artificial Intelligence, Monash University, Australia
[d] Let's Forecast, Australia

ARTICLE INFO

Keywords:
Exponential Smoothing
Bayesian Modelling
Time Series Forecasting
Monte-Carlo Methods
Probabilistic Forecasting

ABSTRACT

This paper describes a family of seasonal and non-seasonal time series models that can be viewed as generalisations of additive and multiplicative exponential smoothing models to model series that grow faster than linear but slower than exponential. Their development is motivated by fast-growing, volatile time series. In particular, our models have a global trend that can smoothly change from additive to multiplicative and is combined with a linear local trend. Seasonality, when used, is multiplicative in our models, and the error is always additive but heteroscedastic and can grow through a parameter sigma. We leverage state-of-the-art Bayesian fitting techniques to fit these models accurately, which are more complex and flexible than standard exponential smoothing models. When applied to the M3 competition data set, our models outperform the best algorithms in the competition and other benchmarks, thus achieving, to the best of our knowledge, the best results of per-series univariate methods on this dataset in the literature. An open-source software package of our method is available.

© 2024 The Author(s). Published by Elsevier B.V. on behalf of International Institute of Forecasters. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

Despite being introduced over 50 years ago, exponential smoothing methods remain among the most widely used methods for forecasting (Goodwin, 2010). Their enduring popularity stems primarily from their relative simplicity, robustness, flexibility, and good forecasting performance (Gardner, 2006). For example, exponential smoothing models were able to outperform several more sophisticated and complex algorithms in the influential M3 forecasting competition (Makridakis & Hibon, 2000), and exponential smoothing models were successfully used as building blocks in many of the methods used in the recent M4 competition (Makridakis, Spiliotis, & Assimakopoulos, 2018).

Exponential smoothing was created as a simple heuristic, in which the next point forecast is given by the previous point forecast plus a correction term that is proportional to the error made by the previous forecast. Over the years, the basic exponential smoothing technique has been extended to include additional aspects such as seasonality (additive and multiplicative), local trends (additive, multiplicative, and damped versions of both), and more complex stochastic error term specification (additive and multiplicative). Exponential smoothing techniques are frequently referred to in the literature by the initialism "ETS", which may stand for either

* Corresponding author at: Department of Computer Science and Artificial Intelligence, University of Granada, Spain.
E-mail address: bergmeir@ugr.es (C. Bergmeir).
URLs: https://cbergmeir.com/ (C. Bergmeir), https://letsforecast.com/ (A. Dokumentov).

https://doi.org/10.1016/j.ijforecast.2024.03.006
0169-2070/© 2024 The Author(s). Published by Elsevier B.V. on behalf of International Institute of Forecasters. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

ExponenTial Smoothing or Error, Trend, and Seasonality. Hyndman, Koehler, Ord, and Snyder (2008), Hyndman, Koehler, Snyder, and Grose (2002) systematically consolidated the various exponential smoothing models and gave them a solid theoretical base by reformulating them within a state-space framework with a single source of error. Subsequently, this theoretical contribution was followed by the creation of the *forecast* package (Hyndman & Khandakar, 2008) in the R programming language (R Core Team, 2018), which implements ETS models in an accessible and user-friendly way within an open-source framework. The package has since become the *de facto* standard tool for forecasting time series in many areas. A more recent continuation of this development is via the *fable* package (O'Hara-Wild et al., 2021), which reimplements and extends the functionality of the *forecast* package. Apart from these main lines of research, other authors have developed various robust versions of exponential smoothing, e.g., (Cipra, 1992; Crevits, Bergmeir, Hyndman, Ihaka, & R Core Team, 2018; Crevits & Croux, 2016; Gelper, Fried, & Croux, 2010). Svetunkov, Kourentzes, and Ord (2022) developed the complex exponential smoothing method, and Sbrana and Silvestrini (2020) present a damped trend model that uses a structural approach, i.e., multiple sources of error formulation.

The ETS models that are currently in use (as described in Hyndman, Koehler, et al. (2008), and implemented in the *forecast* package) are either linear or exponential in their trend, with the only option currently available to potentially bridge this gap being a damped trend. So, the choice is limited, especially if these options cannot suitably model the underlying growth rate. One way to address this shortcoming is by using data transformations as a preprocessing step, such as a logarithm or Box–Cox transformation (Box & Cox, 1964). However, the logarithm is a strong transformation that can lead to explosive behaviour. The strength of the Box–Cox transformation can be adjusted using its parameter λ, but choosing this parameter is not trivial, and we observe in the experiments of our ablation study that standard methods for this task also often lead to explosive forecasts. Attempts have been made in adjacent fields to develop more flexible trend formulations; for example, Parzen (1982) presents a model where the differencing of ARIMA is replaced by other, slightly more complicated non-stationary AR models, thus allowing for more flexible trend choices. Apart from the relatively unflexible trend, the ETS models assume normality of the errors, homoscedasticity of the errors for additive models and have some further shortcomings described in detail in Section 3.

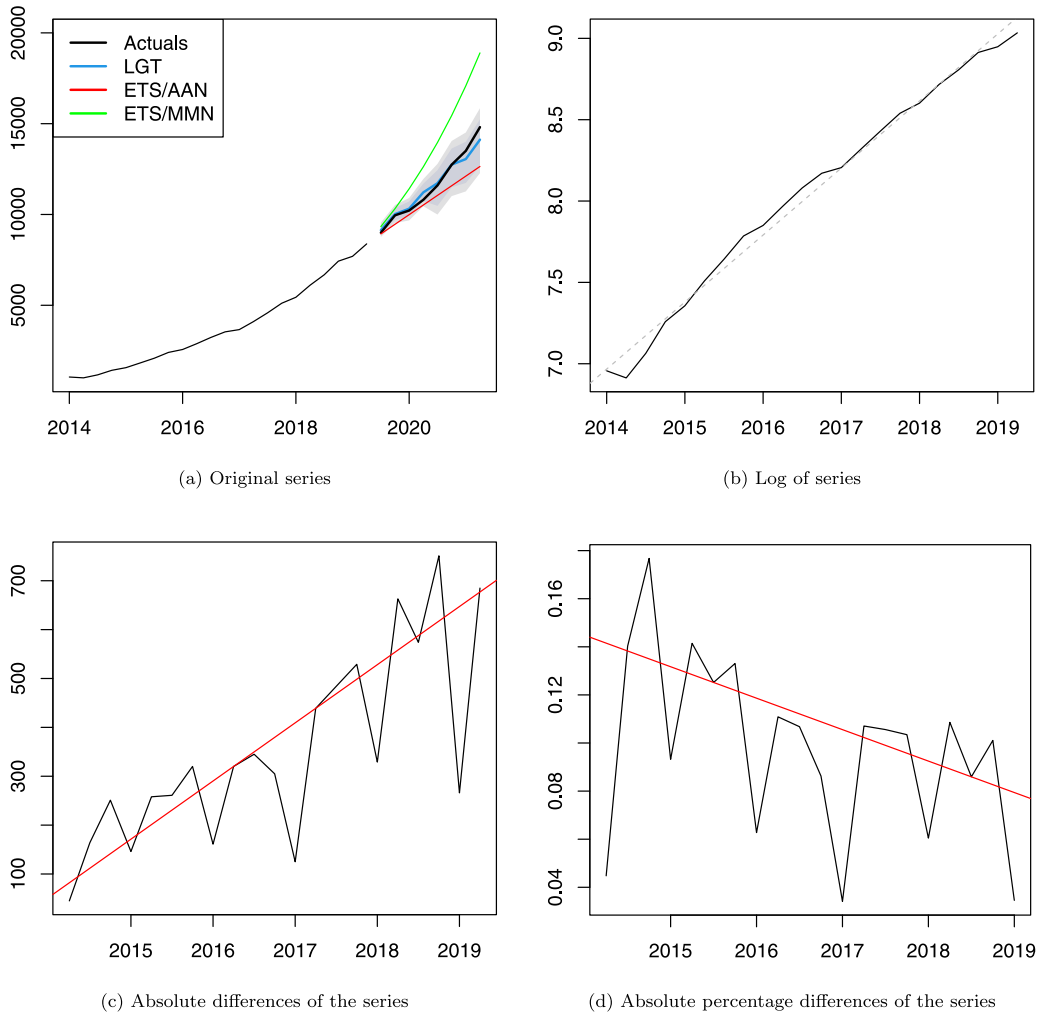
1.1. Bayesian ETS

The majority of development in the ETS space has been through the lens of frequentist statistical approaches. In contrast, applying Bayesian methods to exponential smoothing has been less studied. Important examples of Bayesian ETS approaches are proposed in the works of Andrawis and Atiya (2009) and Bermudez, Segura, and Vercher (2010). Corberan-Vallet, Bermudez, and Vercher (2013) subsequently proposed a related method for time series containing zeros, (Bermudez, Corberán-Vallet, & Vercher, 2009) presented a model suitable for multivariate forecasting, and Corberán-Vallet, Bermúdez, and Vercher (2011) extended the approach to forecasting time series with correlated random disturbances. The work of Bermudez et al. (2010) examined the topic of Bayesian estimation of basic ETS models in a holistic way. The performance of the resulting model was comparable to the best forecasting models in the M3 competition, as well as forecasts produced from popular automatic packages (Bermudez et al., 2010). An interesting result of that study was the demonstration that using Bayesian estimation meant the resulting model could produce accurate prediction intervals close to the empirical ones. This is an important feature that the classical exponential model from the M3 competition is lacking. More recently, Tsionas (2021) extended the work of Sbrana and Silvestrini (2020) to a Bayesian model with multiple sources of error formulation. Interestingly, despite the promising results emerging from the works in this area, the Bayesian approach has not found widespread adoption in the application of ETS models. This could be due to its relatively slow or complicated implementation and a lack of readily available software.

1.2. A motivating example

There are many practical applications, e.g., in cloud infrastructure, rideshare, and other tech businesses, where characteristics of the ETS framework mentioned above do not adequately capture the time series in consideration. In many applications, time series grow faster than linear but slower than exponential, and in situations with volatile time series, the Gaussian error assumption is violated. Strong trends will happen in the real world and eventually slow, e.g., in product lifecycles. This has led to the development of damped trend models in the literature. However, we argue that there are many business use cases where this de-acceleration of the trend may still be far away, and it is unclear when it will happen. Large technological trends, growth in technology governed by Moore's Law, or growth of a highly successful startup company may lead to strongly trending series over decades, and even when they slow down, they may still show growth that is faster than linear. The motivation for our method is to produce short- and mid-term forecasts for such a series.

As an illustrative example, we use the following quarterly revenue from Amazon Web Services, a cloud infrastructure provider. Fig. 1 shows the series and indications that it grows faster than linear but slower than exponential. We also show forecasts of ETS with a linear and exponential trend, and our proposed method, LGT, can forecast the series better. We can also see that the series' error grows slower than proportional to the level. Thus, expressing the scale of the error as a function of level is important in time series with error levels that are one or two orders of magnitude smaller at the beginning than at the end of the series, so homoscedastic models are unsuitable in this case. At the same time, while volatility grows with time series levels, it is seldom proportional to the level.



(a) Original series

(b) Log of series

(c) Absolute differences of the series

(d) Absolute percentage differences of the series

Fig. 1. (a) Quarterly revenue of Amazon Web Services (in million USD, source: statista.com). Forecasts from ETS with linear and exponential trends show that the trend is either under- or over-estimated. LGT can capture the trend adequately. (b) The logarithm of the series confirms that the series grows slower than exponential (as the log of the series doesn't grow linearly). (c) Absolute differences of original data. Assuming errors are bigger than changes in trend, these values estimate absolute errors. Under this assumption, the graph shows that errors grow over time. (d) Absolute differences divided by the first value, i.e., $\frac{|y_{t+1}-y_t|}{y_t}$. We see that errors are not proportional to the trend.

1.3. Our contributions

This paper proposes several practically motivated extensions to the ETS family of models with additive error and additive trend to convert them into flexible nonlinear models. As the models are likely to be analytically unsolvable, and our models use non-Gaussian error distributions, we fit them using a probabilistic programming tool (Stan Development Team, 2015) that employs a fast Markov Chain Monte Carlo engine (Hoffman & Gelman, 2014). In sum, the main contributions of our paper are as follows:

1. Motivated directly by phenomena commonly observed in real-world time series, we implement several extensions of the ETS framework that are designed to increase the flexibility of the base models. In particular, we introduce a novel global trend modelling that can be flexibly adjusted to growth rates that are between linear and exponential.
2. Using Bayesian model fitting procedures, our more flexible models can still be fitted accurately. We can relax the usual Gaussian assumptions and obtain accurate prediction intervals, even for small sample sizes. We implement a standard version of the sampler, rendering the implementation extensible and easy to understand, and a bespoke specialised Gibbs sampler roughly an order of magnitude faster than the standard implementation.
3. This gives us, in sum, an acceptably fast model that can outperform even the most competitive benchmarks in this research area in terms of accuracy. We provide documented and tested open-source software implementations of our proposed methods.

The methods presented in this paper are available in the *Rlgt* package on CRAN.[1] The *Rlgt* package is a comprehensive implementation of models discussed in this paper, with further documentation and illustrations regarding their use.

The remainder of this paper is structured as follows: Section 2 discusses ETS models in detail, Section 3 describes the shortcomings of classical ETS models, and Section 4 discusses Bayesian model fitting and MCMC sampling. Sections 5 and 6 present the non-seasonal and seasonal versions of our new model, respectively; Section 8 describes the experiments and results, and Section 9 concludes the paper.

2. Exponential smoothing

The exponential smoothing algorithm is fundamentally based on the concept of time series decomposition. As discussed in Section 1, ETS performs a decomposition of the time series data into the three main components: (i) stochastic error, (ii) a trend, and (iii) a seasonal component. The trend represents the long-term increase and decrease in the time series and is not necessarily linear over time. The seasonal component represents the perturbation in the time series due to seasonal factors, e.g., quarter of the year or day of the week. Seasonality patterns last for a fixed period of time. Finally, the error component models the random components which the model cannot explain.

These components can interact with each other in either additive or multiplicative ways. Other interactions are also possible in the ETS model, such as damped trends, in which the trend's strength reduces over time (Gardner, 1985). Exponential smoothing methods were proposed as a simple way of generating fast and reliable forecasts for a wide range of time series data in various application areas that do not make too many strong parametric assumptions.

2.1. ETS model family nomenclature

A wide range of different ETS-based models have been introduced over the years. Following the nomenclature in Hyndman and Athanasopoulos (2018), ETS variants commonly model the trend as either additive (A), additive damped (Ad), multiplicative (M), or as having no trend (N). Seasonality is usually modelled as being either additive (A), multiplicative (M), or not present (N), and the error term is usually either additive (A) or multiplicative (M) in nature. The ETS model family constitutes the set of models that can be built from all possible combinations of these choices for error term, trend, and seasonality. For example, an ETS model with additive error, additive damped trend, and multiplicative seasonality, commonly known as the Holt-Winters' damped method, would be denoted as an "AAdM model" under the standard nomenclature. Furthermore, if a component is not further specified, a "Z" is used as a wildcard placeholder for the missing term; for example, an AAZ model is a model with additive error, additive trend, and any form of seasonality.

[1] https://cran.r-project.org/package=Rlgt

2.2. Classical ETS model estimation

The classical ETS models are usually fitted via the method of maximum likelihood. This is the most popular technique for estimating model parameters in classical statistics. This popularity is primarily due to the strong asymptotic properties the maximum likelihood estimates are known to possess (under suitable regularity conditions), such as consistency, asymptotic efficiency, and asymptotic normality (Fahrmeir & Kaufmann, 1985). The method of maximum likelihood is based on the notion of maximising the joint likelihood of the data under the model

$$p(y_1, \ldots, y_n \mid \boldsymbol{\theta}) \equiv p(\mathbf{y} \mid \boldsymbol{\theta}).$$

Here, $\boldsymbol{\theta}$ is a vector of model parameters and $\mathbf{y}$ is a vector of n observed values, i.e., the likelihood function $p(\mathbf{y} \mid \boldsymbol{\theta})$ is defined as the probability of generating a specific set of data $\mathbf{y}$ given the parameter values $\boldsymbol{\theta}$. The estimates of the parameters are defined as the values of the parameters which maximise the likelihood function, i.e.,

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta} \in \Theta}{\operatorname{argmax}} \{p(\boldsymbol{\theta} \mid \mathbf{y})\} \tag{1}$$

Here, $\hat{\boldsymbol{\theta}}$ denotes the maximum likelihood estimate (MLE) and Θ denotes the valid parameter space. For the classical ETS models, the parameter vector $\boldsymbol{\theta}$ usually consists of (a mix of) the following parameters: a damping coefficient, an error variance, initial trend terms, initial seasonality factors, and smoothing factors for the level, trend, and seasonality. For forecasting, it is usual that the estimated parameters are plugged into the relevant equations to produce future predictions (Hyndman & Athanasopoulos, 2018).

2.3. Non-seasonal (damped) trend model

One particular model from the ETS model family, the non-seasonal (damped) linear trend model (AAdN), is given by

$$y_{t+1} \sim N\left(\hat{y}_{t+1}, \sigma^2\right), \tag{2}$$

$$\hat{y}_{t+1} = l_t + \phi b_t, \tag{3}$$

$$l_{t+1} = \alpha y_{t+1} + (1-\alpha)\left(l_t + \phi b_t\right), \tag{4}$$

$$b_{t+1} = \beta\left(l_{t+1} - l_t\right) + (1-\beta)\phi b_t, \tag{5}$$

where y_t denotes the value of the dependent variable of interest at time t, $\hat{y}_{t+1}$ is the conditional expectation of y_{t+1} given the information up to time t, σ denotes the standard deviation of the error distribution, l_t denotes the level at time t, and b_t the local trend at time t. The parameters to be fitted are $\alpha \in (0, 1)$, a smoothing parameter for the level term; $\beta \in (0, 1)$, a smoothing parameter for the local trend term; $\phi \in (0, 1)$, a damping factor; and initial values for level, l_1, and slope, b_1.

Eq. (2) models the distribution of y_{t+1}; specifically, it assumes that the value of y_{t+1} is distributed as per a normal distribution with mean $\hat{y}_{t+1}$ and standard deviation σ. Eq. (3) states that the one-step-ahead prediction value $\hat{y}_{t+1}$ is formed as the sum of the previous level and trend values. Eq. (4) describes the evolution of the level over

time; the estimate of the level at time $t+1$ is given by the weighted average of the observed value y_{t+1} and the previous estimate of the level l_t. The initial level value l_1 is treated as a parameter to be estimated from the data.

Eq. (5) describes the evolution of the trend across the time horizon. Similar to Eq. (4), the updating process of the trend is based on the weighted average of the current trend value $(l_{t+1} - l_t)$ and the previous estimate of the trend b_t. As with the level, the initial value b_1 is treated as a parameter and estimated. The parameter ϕ is a damping factor that reduces the influence of the trend; if $\phi = 1$, the model reduces to the usual AAN (i.e., undamped) model.

We note that when using a linear trend, we assume that in addition to the level changing over time, we expect the trend to change over time. The weighted average estimate for the trend value usually works best if the trend changes gradually over time in the corresponding time series data. More comprehensive explanations of this model can be found in Hyndman and Athanasopoulos (2018).

2.4. Seasonal (damped) trend model

The seasonal counterpart of the non-seasonal (damped) linear trend model introduces a multiplicative seasonal pattern to account for potential seasonality. This model is also known as the Holt-Winters' multiplicative method (Hyndman & Athanasopoulos, 2018).

$$y_{t+1} \sim N\left(\hat{y}_{t+1}, \sigma^2\right) \tag{6}$$

$$\hat{y}_{t+1} = (l_t + \phi b_t) s_{t+1-m}, \tag{7}$$

$$l_{t+1} = \alpha \left(\frac{y_{t+1}}{s_{t+1-m}}\right) + (1-\alpha)\left(l_t + \phi b_t\right), \tag{8}$$

$$b_{t+1} = \beta \left(l_{t+1} - l_t\right) + (1-\beta)\,\phi b_t, \tag{9}$$

$$s_t = \zeta \left(\frac{y_t}{l_t}\right) + (1-\zeta) s_{t-m}. \tag{10}$$

This extension introduces three new quantities: the s_t represents the seasonal factor prevailing at time t, $\zeta \in (0, 1)$ represents a smoothing parameter for the seasonal factors, and $m > 1$ represents the number of seasons comprising a complete period. The main differences between the AAdN model described in Section 2.3 and the seasonal extension are:

- In comparison to Eq. (3), Eq. (7) introduces a multiplicative factor of s_{t+1-m} which estimates the prevailing seasonal factor at time $t+1$. This factor is estimated m periods before (at this point, the seasonal factor is not yet known, so the best estimate we have is the seasonal factor obtained at time $t+1-m$).
- The updating process of the level in Eq. (8) is based on the weighted average of the current level value (note that the data value is modelled as a product of the level and seasonality factor, so the level value is given by y_{t+1}/s_{t+1-m}) and the one-step prediction of the current level value.
- Eq. (10) describes the evolution of the seasonality factor. Similar to the trend and level adjustment process, the seasonality factor is estimated as a weighted average of the current seasonality value y_t/l_t and the most recent estimate of the corresponding seasonality value s_{t-m}.

3. Weaknesses of classical ETS models

As briefly discussed in Section 1, classical ETS models have several shortcomings that can adversely affect their performance in practice. We now describe these weaknesses in more detail.

3.1. Limited functional form of trend

Despite various trends being considered in classical ETS models, the choice may frequently still be too limited; this is illustrated by the motivating example of quarterly AWS revenue, shown in Fig. 1. The trend grows faster than an additive but slower than a multiplicative form. As mentioned earlier, one current way to address this issue is by using a transformation as a preprocessing step, such as a logarithm or a Box–Cox transformation. See, for example, Beaumont (2014). The logarithm is a very strong transform and often leads to explosive behaviour of the forecasts. The Box–Cox transformation has a parameter λ that controls its strength, but it is not trivial to choose. Including its estimation in the MLE has the problem that different values of λ lead to different autocorrelation structures, making it necessary to, e.g., change the ARIMA order during the search (Guerrero, 1993). The method of Guerrero (1993) chooses λ independently of the modelling to stabilise the seasonal variation across the series. While in our motivating example of AWS revenue, the method can achieve a good result similar to our proposed method, in broader experiments, e.g., on the full M3 dataset, the method often leads to explosive behaviour of the forecasts. See the ablation study in the experimental section of our paper for details.

3.2. Non-normal error distribution

Statistical models based on ETS models, both additive or multiplicative, usually assume that the error terms are identically and independently distributed normal random variables. This assumption is fairly restrictive for most applications and is not necessarily appropriate for modelling volatile time series data, such as the example from Fig. 1. The normality assumption implicitly assumes that the probability of obtaining a value that highly deviates from the mean (i.e., an outlier) is extremely low. In some real data, such as the movements of stock prices, there is a consensus that the normality assumption tends to be violated due to high variance, especially in times of crisis. Despite these empirical observations, few works are in the literature to mitigate this issue. Beaumont (2014) addresses the problem with using a Student t-distribution instead of a normal distribution and a Johnson error trend seasonal transform. However, such changes are often not straightforward in the MLE framework, and therefore, the

normality assumption tends to be retained in classical ETS models as even slight deviations from the normality assumption may significantly complicate the likelihood function; this can cause considerable issues as maximisation of the likelihood function is the key step in standard MLE-based procedures. Consequently, the assumption of normality is usually required in classical ETS models, as otherwise, the MLE may be difficult to find, and other inferential quantities derived from the likelihood, such as confidence intervals, may also be very difficult to obtain.

3.3. Heteroscedastic error distribution

In addition to being non-normally distributed, the scale of the errors in time series data is often not homoscedastic, i.e., it may vary over time. This issue has been recognised even in classical ETS models, and various approaches based on innovation state space representations have been introduced to deal with it, with varying levels of success, see, e.g., Hyndman, Koehler, et al. (2008, Ch. 4, 19), and Beaumont (2014). For example, the multiplicative error form in the state space models may allow the error to depend linearly on the current value of the data. Similar to the discussion of the form of the trends in Section 3.1, how the error scale is linked to the trend is usually quite limited and may not be sufficient to model more general time series data accurately. For example, the state-space model might not be able to handle an error function which grows with the data over time (i.e., faster than an additive error form) but at a sub-linear rate (i.e., slower than the multiplicative error form).

4. Bayesian estimation

The overwhelming majority of previous work on exponential smoothing utilises frequentist inferential tools, such as maximum likelihood and confidence intervals, to estimate model parameters and provide measures of uncertainty. Maximum likelihood estimation tends to suffer from high parameter variance, particularly for small sample sizes and probability models with heavy tails, such as the t-distribution; similarly, due to the great difficulty in deriving exact confidence intervals for all but the simplest of models, most uncertainty quantification relies on asymptotic arguments and may provide poor estimates of variability for small sample sizes.

In contrast, we use the Bayesian paradigm of statistical inference to fit our proposed ETS models. In the Bayesian approach, one treats the unknown model parameters, say θ, as realisations of random variables coming from some prior distribution, $\pi(\theta)$, and argues conditionally on the observed data sample, y, to form the posterior distribution $p(\theta \mid y)$:

$$p(\theta \mid y) = \frac{p(y \mid \theta)\pi(\theta)}{p(y)}, \tag{11}$$

where $p(y \mid \theta)$ denotes the likelihood function, and $p(y)$ is the marginal probability of the data under any possible model as a normalising factor. The posterior distribution describes how likely specific values of θ are to be the unknown population value of θ in light of the data and prior beliefs. The Bayesian approach naturally handles complete parameter uncertainty through the mechanism of the prior distribution. This same mechanism also allows us to specify any prior information about the unknown model parameters very transparently and, as such, offers a form of regularisation that can stabilise parameter estimates. In the context of ETS, regularisation has recently been shown to be beneficial in a frequentist approach by Pritularga, Svetunkov, and Kourentzes (2023). An additional strength of the Bayesian approach is that given the posterior distribution over the unknown model parameters, at least in principle, it is straightforward to generate the posterior distribution over any function of the model parameters via standard transformation of random variables techniques. This can be used, for example, to generate posterior distributions over point or distributional forecasts, which naturally incorporate the uncertainty in parameter estimates.

4.1. Monte Carlo Markov chain (MCMC) methods

Despite the attractive theoretical advantages of the Bayesian approach, the primary weakness is the potential implementation difficulties. Computation of the normalising factor $p(y)$ in (11) involves high-dimensional integration and is, in general, intractable. Even if one could compute it, effectively exploring an arbitrary multidimensional probability distribution is problematic. Instead, one generally resorts to some form of approximation; one of the most popular approximation techniques is simulating values of θ from the posterior distribution. This is usually done via Monte Carlo Markov Chain (MCMC) techniques that do not require knowledge of the normalising term and are thus very convenient. Given a random sample of m values drawn from (11), say $\theta^{(1)}, \ldots, \theta^{(m)}$, it is straightforward to compute quantities such as the posterior mean, standard deviation and intervals using the corresponding empirical (sample) quantities. Furthermore, if we have some function of the model parameters, $f(\theta)$, we can approximate the posterior distribution of this function by using

$$f\left(\theta^{(1)}\right), \ldots, f\left(\theta^{(m)}\right),$$

i.e., we simply evaluate $f(\cdot)$ for each sample $\theta^{(i)}$ we have drawn from the posterior distribution. In the context of this paper, the typical functions we may evaluate would be quantities such as forecasts over future data.

4.2. Probabilistic programming

The application of Bayesian statistical techniques to practical data science problems has increased dramatically in the last two decades. This has primarily been driven by increases in computational power, coupled with the introduction of easy-to-use Bayesian probabilistic programming tools such as Stan (Stan Development Team, 2015) and JAGS (Plummer, 2003). These tools enable non-specialist data scientists to perform Bayesian modelling and computation more efficiently, specifically in terms of development time and programming effort. Stan is a probabilistic programming language that is focused on

Bayesian statistical modelling. The underlying computational algorithm used by Stan to sample from the posterior distribution is a variation of the Hamiltonian MCMC algorithm (Neal, 1994), known as the "No U-Turn Sampler" (NUTS). This procedure uses autodifferentiation to automatically build an MCMC sampler from the posterior distribution specified by the user. Conveniently, Stan is also integrated into R through the package *Rstan* (Stan Development Team, 2018).

Although Stan offers several advantages as a state-of-the-art Bayesian tool, the primary benefit of the tool stems from the separation between modelling (through the probabilistic programming language) and computational processes (Betancourt, 2017) (through the built-in Hamiltonian sampling algorithm) that it provides. This separation enables users to focus primarily on the specification of the problem, i.e., the choice of model and prior distributions, rather than the computational aspects of sampling. For most problems, users are not required to specify any computational procedures, as they can be handled adequately by the default process. However, the algorithm is not fool-proof, and in general, both run-time and convergence of the resulting Hamiltonian sampler can be heavily influenced by how models and priors are specified.

In addition to Stan, several other probabilistic programming tools support Bayesian automated MCMC sampling. In particular, JAGS (and its predecessor BUGS) has also been extensively used by Bayesian data analysts (Lunn, Thomas, Best, & Spiegelhalter, 2000). In comparison to Stan, JAGS uses an automatic Gibbs sampling approach which can be less efficient than the NUTS algorithm used by Stan (R-bloggers, 2014).

5. Local and global trend non-seasonal model

To address the shortcomings identified in Section 3, we propose the non-seasonal local and global trend model (LGT, note that "local" and "global" refer here to parts of the series or the whole series, as opposed to models that learn across time series and are also often called global models, see, e.g., Januschowski et al. (2020)). It is derived from the classical ETS model with additive error, additive trend, and no seasonality, with the addition of several distinguishing features. We define it as follows.

$$y_{t+1} \sim t(\nu, \hat{y}_{t+1}, \hat{\sigma}_{t+1}), \tag{12}$$

$$\hat{y}_{t+1} = l_t + \gamma l_t^{\rho} + \lambda b_t, \tag{13}$$

$$l_{t+1} = \alpha y_{t+1} + (1-\alpha)\, l_t, \tag{14}$$

$$b_{t+1} = \beta\,(l_{t+1} - l_t) + (1-\beta)\, b_t, \tag{15}$$

$$\hat{\sigma}_{t+1} = \sigma \hat{y}_{t+1}^{\tau} + \xi. \tag{16}$$

Here, $t(\nu, m, s)$ denotes a Student t-distribution with degrees-of-freedom ν, location m and scale s. Furthermore, y_t denotes the value of the dependent variable of interest at time t, $\hat{y}_{t+1}$ is the predicted value (conditional mode) of y at time $t+1$ given information up to time t; $\hat{\sigma}_{t+1}$ denotes the scale of errors at time $t+1$; l_t is the level at time t; and b_t the local trend at time t. The parameters that must be estimated are described in Table 1.

Table 1
Parameters to be fitted for the LGT model.

	Description
ν	degrees of freedom of the t-distribution
γ	coefficient of the global trend
ρ	power coefficient of the global trend, by default in $[-0.5, 1]$
λ	damping coefficient of the local trend, in $[0, 1]$
α	smoothing parameter for the level term, in $[0, 1]$
β	smoothing parameter for the local trend term, in $[0, 1]$
σ	coefficient of the size of the error, positive
τ	power coefficient of the size of error, in $[0, 1]$
ξ	minimum value of the size of the error, positive
b_1	initial local trend

Eq. (12) models observation y_{t+1} by a Student t-distribution, where Eq. (13) defines the conditional mode (i.e., the predicted value) of the Student t-distribution at the next time-step. Eq. (14) estimates the level adjustment via exponential smoothing, and Eq. (15) estimates the local trend via exponential smoothing of the first differences of the smoothed levels, and Eq. (16) defines the level dependent (heteroscedastic) scale of the error distribution.

Eq. (14) is reminiscent of the Holt linear method with a slightly simpler structure. The local trend update formula is the same as in the Holt linear (AAN) method; see Eq. (5). The main differences are now discussed in more detail.

5.1. Student t-distributed errors

Eq. (12) models the observations via a Student t-distribution with a time-varying scale. We note that the Student t-distribution has three free parameters as opposed to the two free parameters of the normal distribution. The additional parameter, ν, commonly referred to as the degrees-of-freedom, controls the weight of the tails of the distribution, which facilitates the graceful handling of "outliers" in data. This choice of data distribution is in contrast with the innovation state space models for classical ETS (Hyndman & Athanasopoulos, 2018), in which the error distribution is assumed to be normal and identically distributed with a mean of zero and variance of σ^2.

However, this assumption is often overly restrictive, as outlined in Section 3. This motivates our choice to generalise the error assumption from normal to Student t. The Student t-distribution is a generalisation of the normal distribution, with the heaviness of the tails being smoothly controlled by the degrees-of-freedom parameter ν. When ν is relatively high (i.e., above 20), the Student t is virtually identical to a normal distribution. In contrast, when ν is small, the tails of the t-distribution are heavier than the tails of a normal distribution; when $\nu = 1$, the t-distribution reduces to the special case of the Cauchy distribution.

Like all other model parameters, ν is fitted automatically within our Bayesian approach. This means the model can automatically adapt to relatively smooth time series without large spikes (e.g., like those of the M3 competition) but can handle the more volatile series often found in practical applications.

We also note that, in principle, our models can deal with asymmetric distributions. However, the results of preliminary experiments indicated that the fitting was slower and results were not improved, so we did not pursue this option further.

5.2. One-step ahead prediction

In the classical AAN model (Eq. (3)), the forecasts are calculated as $\hat{y}_{t+1} = l_t + b_t$, where l_t is the current estimate of the level and b_t is the current estimate of the trend, resulting in a linearly changing forecast. In multiplicative models the one-step-ahead prediction is instead given by $\hat{y}_{t+1} = l_t\, b_t$ (Hyndman & Athanasopoulos, 2018), which results in an exponentially varying forecast. In the latter case, for a model to be reasonably stable, the trend values b_t need to be close to one. Define the trend to be $b_t = 1 + \delta_t$, where δ_t is a small value; then $\hat{y}_{t+1} = l_t b_t = l_t\,(1 + \delta_t) = l_t + l_t\delta_t$. This suggests that both the additive and multiplicative case can be generalised by the following equation:

$$\hat{y}_{t+1} = l_t + b_t l_t^{\rho}, \tag{17}$$

with $\rho \in [0, 1]$. While this generalisation of the additive and multiplicative trend model is interesting, we found in practice that its performance was typically worse than that of the alternative global trend model:

$$\hat{y}_{t+1} = l_t + \gamma l_t^{\rho}, \tag{18}$$

where γ is constant for the whole time series. This flexible form of trend function is particularly useful for dealing with time series that grow faster than linear but slower than exponential. If the ρ parameter is allowed to be negative, then for a growing trend, we can obtain a damped behaviour. In practice, e.g., when fitting to the M3 time series, we did not observe this capability as particularly important.

Additionally, the model includes a local linear trend of the form λb_t, where λ is a value between -1 and 1. In practice, this trend is almost always between 0 and 1. Restricting the magnitude of this coefficient means that the model is sensitive to the recent changes in slope but to a lesser extent than the original AAN model. A negative λ coefficient indicates a counter-trending behaviour, which is relatively rare. The one-step-ahead forecast is then:

$$\hat{y}_{t+1} = l_t + \gamma l_t^{\rho} + \lambda b_t. \tag{19}$$

It is assumed, and enforced during the simulation (i.e., forecasting) stage, that the level and forecasting values are all strictly positive and larger than 0.001 to avoid numerical instabilities. All model parameters, such as γ, ρ, and λ, are fitted via MCMC sampling from the posterior distribution.

The one-step-ahead forecast given by Eq. (19) is provided by the linear combination of the level l_t, the global nonlinear trend γl_t^{ρ}, and the damped local linear trend λb_t. We use the term "global" to describe the nonlinear trend to stress that both γ and ρ are fitted on the entire time series and do not change over time. The simple function that defines the global trend has several interesting properties. If ρ is close to zero, the contribution of the global trend to the one-step-ahead prediction becomes near-constant (a drift term), and the resulting trend becomes close to linear, leading to an overall (global) upward or downward trend. If ρ is not zero, the global trend is a time-invariant function of the level. The level is typically smooth but puts more weight on recent observations; one could argue that there is a local component in the "global" trend. However, the global trend would be smooth and change slowly, whereas the local trend component can change much quicker. Typically, the global trend will eventually dominate over the short-term behaviour caused by the local linear trend. This combination of local and global trends can be especially useful, for example, in situations like the COVID-19 pandemic, as the model can simulate both long-term overall trends and short-term changes in the behaviour of the series. Finally, if ρ is close to one, the contribution of the global trend to the one-step-ahead prediction becomes almost proportional to the level value, and the trajectory of the multistep forecast becomes exponential. This means that the global trend is flexible enough to adequately model trends typical of fast-growing businesses, i.e., growth that is faster than linear but slower than exponential.

5.3. Heteroscedastic error

In addition to accounting for a possible heavy-tailed error distribution, our proposed model also allows the error scale to vary as a function of the estimated time series level. This is achieved by allowing the scale parameter of the Student's t-distribution to vary according to a monotonically increasing function of the current time series level. This is given by Eq. (16), which models the error scale by a power transformation of the one-step-ahead prediction plus a level-independent base scale term ξ.

This allows us to account for the common situation in which the magnitude of the error is an increasing function of the value of the time series, with the use of a variable power τ allowing us to moderate the rate at which the scale increases as the value of the time series increases. In practice, τ is usually limited to values between zero and one. A τ of zero corresponds to the situation of a constant scale across the series, i.e., homoscedasticity. In contrast, $\tau = 1$ approximates the behaviour of the multiplicative error ETS model, in which the magnitude of the error is proportional to the expected value. In practice, a value between zero and one is desirable for τ as we expect the error to grow in relation to the magnitude of the time series, but in a sub-linear manner. The parameter $\xi > 0$ sets the base error scale. These modifications relax the usual NID assumption, as the error term is now not required to be normally or identically distributed.

6. Seasonal global trend model

We also consider the extension of our procedure to seasonal models. Specifically, we introduce the seasonal global trend model (SGT), a seasonal modification of the previously discussed LGT model. In contrast to the non-seasonal model, we remove the local linear trend term

and replace it with the ability to model seasonal effects. Though the local linear trend could be maintained, in preliminary experiments not reported here, we did not find a benefit from having the local trend, and fitting such models was slower. We hypothesise that the models get too complex with too many parameters. We utilise a multiplicative seasonality scheme, with the seasonality coefficients evolving as per the usual ETS model:

$$l_t = \alpha \cdot \frac{y_t}{s_t} + (1-\alpha) \cdot l_{t-1} \tag{20}$$

$$s_{t+m} = \zeta \cdot \frac{y_t}{l_t} + (1-\zeta) \cdot s_t \tag{21}$$

Here, $m > 0$ is the seasonality (e.g., $m = 12$ for a monthly series), $\zeta \in [0, 1]$ is the seasonality smoothing coefficient, and s_t are seasonality factors.

A key difference in our Bayesian implementation, *vis à vis* the conventional seasonal ETS model, is how the initial seasonality coefficients, $s_1, \ldots, s_m$, are handled. Rather than calculate the ratio of observations against a cyclical mean for a few cycles, potentially followed by maximum likelihood estimation, they are handled in a fully Bayesian fashion. That is, we simply treat these quantities as unknown parameters and assign them appropriate prior distributions, specifically $\log s_i \sim C(0, b)$, where $C(a, b)$ denotes a Cauchy distribution with location a and scale b. We then allow these quantities to be sampled along with the other model parameters as part of the MCMC sampling procedure.

The complete specification of the model is:

$$y_{t+1} \sim t(\nu, \hat{y}_{t+1}, \hat{\sigma}_{t+1}) \tag{22}$$

$$\hat{y}_{t+1} = (l_t + \gamma l_t^{\rho}) s_{t+1} \tag{23}$$

$$l_{t+1} = \alpha \frac{y_{t+1}}{s_{t+1}} + (1-\alpha)\, l_t \tag{24}$$

$$s_{t+m} = \zeta \cdot \frac{y_t}{l_t} + (1-\zeta) \cdot s_t \tag{25}$$

$$\hat{\sigma}_{t+1} = \sigma \hat{y}_{t+1}^{\tau} + \xi \tag{26}$$

with

$$\frac{1}{m} \sum_{i=1}^{m} s_i = 1.$$

The latter condition requires that the seasonality adjustments be expressed relative to the overall scale of the signal. The complete set of parameters that need to be fitted for the SGT model are described in Table 2.

During the forecasting (simulation) phase, the seasonal coefficients are not updated using a frozen setup of the last m factors. The motivation for this process is that these coefficients are meant to adjust to changing situations (new data), but during the forecasting phase, we do not have new data available.

7. Implementation

In Sections 5 and 6, we introduced several extensions to the classical ETS model, which allowed for a wider range of trend and error models. We fit these models within the Bayesian framework of statistical inference. Specifically, we use MCMC approaches to simulate the posterior distribution and use these to approximate the posterior predictive distribution over forecasts. We provide two implementations: (i) a Stan-based implementation and (ii) a bespoke Gibbs sampling implementation based on a simplified model. The former is more flexible and easier for users to extend, but it is slow. The Gibbs sampler is less straightforward to extend but is substantially faster and does not require a working Stan installation to run. In the following, we detail the approaches.

Table 2
Parameters to be fitted for the SGT model.

	Description
ν	degrees of freedom of the t-distribution
γ	coefficient of the global trend
ρ	power coefficient of the global trend, by default in $[-0.5, 1]$
α	smoothing parameter for the level term, in $[0, 1]$
ζ	smoothing parameter for the seasonality terms in $[0, 1]$
σ	coefficient of the size of the error, positive
τ	power coefficient of the size of the error, in $[0, 1]$
ξ	minimum value of the size of the error, positive
s_i	initial seasonality, positive; for i in $\{1, \ldots, m\}$

7.1. Prior distributions and parameter space

A key step in Bayesian inference is the specification of prior distributions, from which the model parameters are randomly sampled. The choice of prior distributions can substantially affect the performance of a Bayesian procedure, particularly for small sample sizes, and their parameters parametrise a number of these prior distributions so that a user can influence the initialisation and fitting. In our case, the random sampling from the prior distributions replaces the relatively sophisticated initial choices for ETS model parameters in traditional ETS models.

Also, the bounds within which the parameters can be set have received considerable attention in the literature for ETS models (Hyndman, Akram, & Archibald, 2008; Hyndman & Athanasopoulos, 2018; Hyndman, Koehler, et al., 2008). The parameters are traditionally restricted to the intervals $0 < \alpha < 1$, $0 < \beta < \alpha$, and $0 < \zeta < 1-\alpha$ (the seasonal parameter is called γ in that literature, but we use ζ in our paper). The damping parameter ϕ is constrained, e.g., to $0.8 < \phi < 0.98$, to prevent numerical problems (Hyndman & Athanasopoulos, 2018). Furthermore, there are bounds for forecastability and stability conditions. For the additive ETS model variants we focus on, these are less tight than the traditional restrictions (Hyndman, Akram, & Archibald, 2008; Hyndman & Athanasopoulos, 2018), so they are not relevant to our case. Finally, for the seasonal models, ETS (in its implementation in the *forecast* package) also checks that the roots of the characteristic equation all lie within the unit circle (also see Hyndman, Akram, and Archibald (2008)).

As our models have several modifications compared to the traditional ETS models, other parameter region

constraints may be applicable. However, we do not perform a detailed theoretical study of these bounds; instead, we choose bounds that align with traditional ETS and that empirically lead to good forecasts. In particular, we follow Bermudez et al. (2010), who have presented a Bayesian exponential smoothing approach for an additive Holt-Winters (AAA) model. Those authors constrain the smoothing parameters in the interval (0,1) and use non-informative priors.

We use the following priors in our implementation. The degrees-of-freedom ν and power-coefficients ρ and τ are assigned uniform distributions over appropriate intervals; the scale σ and minimum error scale ξ are assigned half-Cauchy distributions; the smoothing parameters α, β and ζ (for seasonal models) are assigned beta distributions; the coefficients γ and λ are assigned heavy-tailed, weakly informative Cauchy distributions (Gelman, 2006); and the initial values for the seasonality terms, $s_1, \ldots, s_m$ are assigned log-Cauchy prior distributions. The hyperparameters for these prior distributions are given default values in our implementation, and there is, in principle, no need for the user to specify these.

However, the choice of priors can significantly influence the model outcomes, and the user might have prior knowledge to be incorporated into the modelling in the form of a tightened prior. Due to the flexibility of Stan, it is straightforward for the user to modify these prior hyperparameter values if so desired.

7.2. Gibbs sampler

In addition to providing a Stan implementation, we also developed an MCMC sampling algorithm for our proposed model via a bespoke Gibbs sampler.

7.2.1. Modified model

For purposes of tractability, the Gibbs sampler is based on a slightly simplified form of the LGT model:

$$\begin{aligned} y_{t+1} \mid \hat{y}_{t+1}, \sigma, \nu &\sim t(\nu, \hat{y}_{t+1}, \sigma), \\ \hat{y}_{t+1} &= l_t + \gamma l_t^{\rho} + \lambda b_t, \\ l_t &= \alpha y_t + (1-\alpha) l_{t-1}, \\ b_t &= \beta(l_t - l_{t-1}) + (1-\beta) b_{t-1}. \end{aligned}$$

The simplified model enforces homoscedasticity on the errors, which dramatically improves the convergence speed of the resulting Gibbs sampler. For the purposes of sampling, the Gibbs sampler represents the t-distribution via a scale-mixture of normal distributions (Lange, Little, & Taylor, 1989), i.e.,

$$\begin{aligned} y_{t+1} \mid \mu_t, \sigma, \omega_{t+1}^2 &\sim N(\mu_t, \sigma^2 \omega_{t+1}^2), \\ \omega_{t+1}^2 \mid \nu &\sim \mathrm{IG}\left(\frac{\nu}{2}, \frac{\nu}{2}\right), \end{aligned}$$

where $\mathrm{IG}(a, b)$ denotes the inverse-gamma distribution. This representation is equivalent, as integrating out the latent variables ω_t^2 exactly recovers the t-distribution. Despite introducing n new latent variables that need to be sampled, this representation lets us treat $\hat{y}_t$ as the linear predictor in a regular Gaussian linear regression, making sampling γ and λ substantially easier. We also utilised this same scale-mixture form to represent the Cauchy prior distributions and an inverse-gamma scale mixture of inverse-gammas to represent the half-Cauchy prior distribution (Wand, Ormerod, Padoan, & Fruhwirth, 2011).

A second difference between the Gibbs sampler and the Stan implementation is in the way in which the ρ and ν parameters are handled. Given the non-standard form of the conditional distributions for these parameters, the Gibbs sampler utilises a simple grid-sampler by restricting the range of possible values these parameters can take to a finite, discrete set. We found this has little effect on the performance of the methods, as the models appear insensitive to the specific values of these parameters but substantially improve convergence of the resulting algorithm.

7.2.2. The sampling algorithm

We now briefly summarise the Gibbs sampling algorithm. Gibbs sampling from a posterior distribution, say $p(\theta_1, \theta_2 \mid y)$ works by iteratively sampling from the conditional distributions $\theta_1 \mid \theta_2, y$ and $\theta_2 \mid \theta_1, y$. This can be efficient when it is relatively easy to draw random samples from the conditional distributions. Let $\hat{y} = (\hat{y}_1, \ldots, \hat{y}_n)$ denote the vector of one-step-ahead forecasts made by the LGT model, and $\omega = (\omega_1, \ldots, \omega_n)$ denote the vector of latent variables. We now summarise the specific Gibbs sampling steps:

1. Sample $\sigma^2 \mid y, \hat{y}, \omega$. Due to using the scale-mixture representation for the t-distribution the half-Cauchy prior, this conditional distribution is inverse-gamma.
2. Sample the latent variables $\omega \mid y, \hat{y}, \nu, \sigma^2$; the conditional distributions for these are also of the inverse-gamma type (Lange et al., 1989).
3. Sample $\nu \mid \omega$ using a grid-sampler.
4. Sample the coefficients $\gamma, \lambda \mid y, \hat{y}, \omega, \sigma^2$; due to the latent-variable representation of the likelihood and prior distributions, the conditional distributions are just (truncated) normal distributions.
5. Sample $\alpha, \beta \mid y, \hat{y}, \nu, \sigma^2$ using an adaptive Metropolis–Hastings sampler (Schmidt & Makalic, 2019). For this step, we integrate the latent variables ω out of the likelihood to improve the convergence of the Gibbs sampler.
6. Sample the power coefficient $\rho \mid \hat{y}, \nu, \sigma^2$ using a grid-sampler, again integrating out the latent variables ω.

These steps are iterated repeatedly until the desired number of samples has been generated.

8. Experiments

As our proposed models are extensions of the traditional exponential smoothing model family, they are per-series univariate models and do not train across series or utilise additional covariates. As such, we opt to evaluate the classical M3 benchmark dataset. This has been the standard benchmark of datasets in forecasting

Table 3
Dataset characteristics of the M3 dataset. Here, f is the frequency, $\#s$ is the amount of time series, l_{min} and l_{max} are the minimum and maximum length of the series, and h is the forecast horizon.

f	$\#s$	l_{min}	l_{max}	h
Yearly	645	20	47	6
Quarterly	756	24	72	8
Monthly	1428	66	144	18
Other	174	71	104	8
Total	3003			

for decades and remains an excellent and highly representative dataset for evaluating univariate methods of the type presented in this paper. Thus, the M3 dataset is the focus of our experiments.

We use the M3 time series and forecasts from the "*Mcomp*" R package (Hyndman, Akram, & Bergmeir, 2013). The M3 dataset consists of 3003 time series in the yearly, quarterly, monthly, and other categories. Details of the dataset are given in Table 3. The quarterly and monthly categories are seasonal at different levels of seasonality, and the remaining two are non-seasonal.

In the absence of a more sophisticated model selection mechanism, as, e.g., ETS has it (and that could further improve the results of our model), for the two categories that are assumed to be seasonal, we use the SGT model, whereas for the non-seasonal categories we use the LGT model.

8.1. Error measures

When evaluating our new algorithms in terms of point forecasting, we use the following two error measures: (i) the symmetric mean absolute percentage error (sMAPE), which was the primary measure used in the M3 competition (Makridakis & Hibon, 2000), given by Eq. (27), and (ii) the mean absolute scaled error (MASE, Hyndman & Koehler, 2006), given by Eq. (28).

$$\text{sMAPE} = \frac{200}{h}\sum_{t=1}^{h}\frac{|y_{n+t} - \hat{y}_{n+t}|}{|y_{n+t}| + |\hat{y}_{n+t}|}, \tag{27}$$

$$\text{MASE} = \frac{h^{-1}\sum_{t=1}^{h}|y_{n+t} - \hat{y}_{n+t}|}{(n-s)^{-1}\sum_{t=s+1}^{n}|y_t - y_{t-s}|}. \tag{28}$$

Here, h is the maximum prediction horizon, s is the seasonality ($s = 1$ for non-seasonal series and $s > 1$ for seasonal series, e.g., $s = 4$ for quarterly series), and n is the length of the time series.

The motivation to use these two measures is as follows. We use sMAPE as it is the error measure used in the original M3 competition to ensure that the results are directly comparable and avoid any irregularities reported in the literature (Boylan, Babai, Mohammadipour, & Syntetos, 2015). Even though sMAPE has well-documented problems (see, e.g., Hyndman and Koehler (2006)), we argue that it may be the fairest metric to compare with the original participants, as they would have optimised their methods towards this metric. The MASE is used as a measure that is accepted nowadays in the forecasting community as a measure that does not have such problems.

For probabilistic forecasting, we use the Mean Scaled Interval Score (MSIS) as defined in Eq. (29), designed to evaluate prediction intervals. Gneiting and Raftery (2007) first proposed the interval score that computes the sum over the size of the intervals and the magnitude of error for points outside of the interval. The MSIS, as used by Makridakis et al. (2018), which we follow here, achieves a scale-free measure by dividing by the in-sample naïve forecast in the same way as the MASE. In Eq. (29), $q_t^{[u]}$ and $q_t^{[l]}$ are the upper and lower bounds of the prediction interval, respectively, $\mathbb{1}$ is the indicator function that has a value of one if the condition is true, zero otherwise, and α is the significance level, i.e., it defines the target width of the interval. For example, for 90% prediction intervals, we set α to 0.1.

$$\text{MSIS} = \frac{h^{-1}\sum_{t=n+1}^{n+h}\left(q_t^{[u]} - q_t^{[l]} + \frac{2}{\alpha}(q_t^{[l]} - y_t)\mathbb{1}_{y_t < q_t^{[l]}} + \frac{2}{\alpha}(y_t - q_t^{[u]})\mathbb{1}_{y_t > q_t^{[u]}}\right)}{(n-s)^{-1}\sum_{t=s+1}^{n}|y_t - y_{t-s}|} \tag{29}$$

8.2. Statistical testing

We use statistical testing and critical difference diagrams as proposed by Demšar (2006) and later extended by Benavoli, Corani, and Mangili (2016) in a Python implementation of Bunse (2023). In particular, we use a Friedman test to find differences in the outcomes and a Wilcoxon signed-rank test with Holm adjustment for multiple comparisons to characterise the differences. The statistical significance results are visualised in a critical difference diagram where a black line connecting the methods means that no statistical significance between their results could be determined.

8.3. Results and comparison

To facilitate a fair assessment of our proposed univariate methods, we restrict the scope of competitor methods to per-series univariate techniques, excluding any methods that utilise across-series information from our comparison. This is consistent with an application case when only a single time series or a small amount of series can be forecasted. Thus, no across-series modelling is possible.

Table 4 displays the performance of point forecasting for several forecasting algorithms on the M3 time series, averaged over all series belonging to a particular category. The non-seasonal LGT and seasonal SGT models proposed in this paper are run using the Stan sampler. We run LGT on the non-seasonal series, i.e., the yearly and other categories, and SGT on the seasonal series, i.e., quarterly and monthly series. When reporting overall results, we call this procedure {L,S}GT. Furthermore, we report the best per-category/per-metric algorithms from the original M3 competition participants. The only papers that we are aware of that have claimed to achieve some level of improvement over the original M3 competition participants, without cross-learning, are baggedETS as proposed by Bergmeir, Hyndman, and Benítez (2016) and the Multi Aggregation Prediction Algorithm (MAPA) presented in Kourentzes, Petropoulos, and Trapero (2014).

Table 4
Overall point prediction performance on the M3 dataset.

	sMAPE	MASE
All (amount of series: 3003)		
{L,S}GT	**12.29**, sd=0.010	**1.30**, sd=0.0007
Best algorithm in M3	12.76 (THETA)	1.39 (THETA)
Hybrid	12.82	1.40
ETS/ZZZ	13.07	1.43
ARIMA	13.58	1.46
MAPA	12.71	1.41
THETA	12.79	1.42
baggedETS	12.95	1.42
Yearly (amount of series: 645, horizon: 6)		
LGT	**15.18**, sd= 0.015	**2.48**, sd= 0.004
Best algorithm in M3	16.42 (RBF)	2.63 (ROBUST-Trend)
Best of baggedETS (orig)	17.80 (BLD.Sieve)	3.15 (BLD.MBB)
Hybrid	16.73	2.85
ETS/ZZZ	17.00	2.86
ARIMA	17.12	2.96
THETA	16.75	2.77
MAPA	16.91	2.86
baggedETS	17.18	2.88
Other (amount of series: 174, horizon: 8)		
LGT	**4.25**, sd=0.017	**1.72**, sd=0.002
Best algorithm in M3	4.38 (ARARMA)	1.86 (AutoBox2)
Hybrid	4.33	1.79
ETS/ZZZ	4.37	1.81
ARIMA	4.46	1.83
THETA	4.92	2.27
MAPA	4.48	1.87
baggedETS	4.72	1.96
Monthly (amount of series: 1428, horizon: 18)		
SGT	13.77, sd=0.019	**0.83**, sd=0.0006
Best algorithm in M3	13.89 (THETA)	0.85 (ForecastPro)
Best of baggedETS (orig)	13.64 (BLD.MBB)	0.85 (BLD.MBB)
Hybrid	13.91	0.84
ETS/ZZZ	14.14	0.86
ARIMA	14.98	0.88
THETA	13.86	0.86
MAPA	**13.62**	0.84
baggedETS	13.71	**0.83**
Quarterly (amount of series: 756, horizon: 8)		
SGT	**8.87**, sd=0.011	**1.07**, sd=0.0019
Best algorithm in M3	8.96 (THETA)	1.09 (THETA)
Best of baggedETS (orig)	10.03 (BLD.Sieve)	1.22 (BLD.MBB)
Hybrid	9.39	1.14
ETS/ZZZ	9.68	1.17
ARIMA	10.00	1.19
THETA	9.20	1.11
MAPA	9.31	1.13
baggedETS	9.81	1.16

Thus, we also include the sMAPE and MASE scores for the best variants of bagged algorithms, as reported in the yearly, quarterly, and monthly series in Bergmeir et al. (2016). However, that paper doesn't provide the *Other* category results. Regarding MAPA, the sMAPE metric used by those authors is the same as Eq. (27), whereas they use a different definition of the MASE; those authors always use $s = 1$ while differentiating in-sample data. However, their results reported on the benchmarks are quite different. For example, sMAPE of auto ETS of the yearly time series is reported as 18.7, while we calculate it as 17.00. An explanation could be changes/improvements in the default parameter configuration of auto ETS in the last years. Due to these discrepancies, we include results for MAPA of an experimental run that we perform that consistently shows better results than what is reported in the original paper. Furthermore, we also include runs of the standard auto ETS, auto ARIMA, and THETA techniques from the R *forecast* package (Hyndman & Khandakar, 2008), and the mean of the auto ETS and auto ARIMA, which is denoted by "hybrid" in our table. Finally, we also run a version of baggedETS ourselves by running the baggedETS function from the *forecast* package with default parameters.

Table 4 shows that the LGT/SGT algorithms are superior to their competitors in terms of the sMAPE and MASE

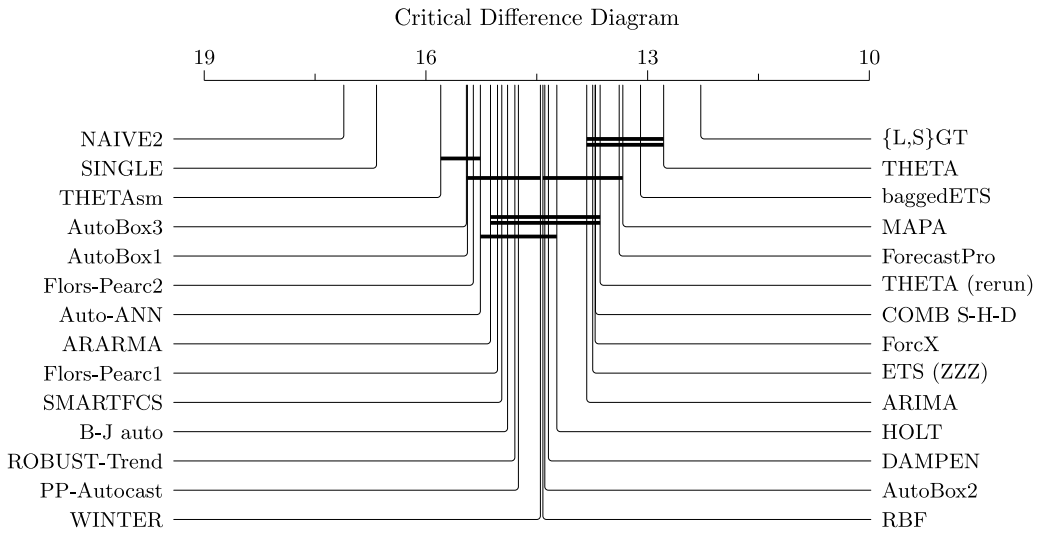


Fig. 2. Results of the statistical testing in a critical difference diagram. The *x*-axis is the average rank across time series in terms of MASE performance. Black bars show methods that are not statistically significantly different from each other. We see that {L,S}GT obtains the best average rank and is statistically significantly better than all comparison methods.

Table 5
Average and Maximum of absolute deviation in per cent of forecasted values across 7 runs.

Series	Avg [%]	Max [%]
Yearly	0.43	50.1
Other	0.11	5.9
Monthly	0.39	89.6
Quarterly	0.17	9.1
All	0.35	89.6

metrics in almost all cases, except sMAPE for monthly time series, for which they still outperform all participants in the original M3.

Fig. 2 shows a critical difference diagram with the results of statistical testing. Methods are ranked per series on all series of the M3 dataset according to their performance on MASE, and then statistical testing is applied. We see that {L,S}GT outperforms with statistical significance all comparison methods, which are our runs of benchmarks as before, together with all original participants in the M3, for which per-series forecasts are available so that they can be included in a per-series rank calculation.

Finally, it is important to note that as the proposed LGT/SGT models are fitted with MCMC (i.e., sampling from the posterior), each run yields slightly different results. However, the standard deviations of metrics also shown in Table 4 (e.g., 0.0006 for Monthly series' MASE) are comparatively small, demonstrating that the performance is quite stable. Furthermore, Table 5 shows average and maximum absolute differences across all forecasted values in all series across seven runs of the method. We see some cases where these deviations are large (the maxima). This happens when the true value is small. However, the average deviations are small, e.g., below 0.5%, so the method is usually stable across runs.

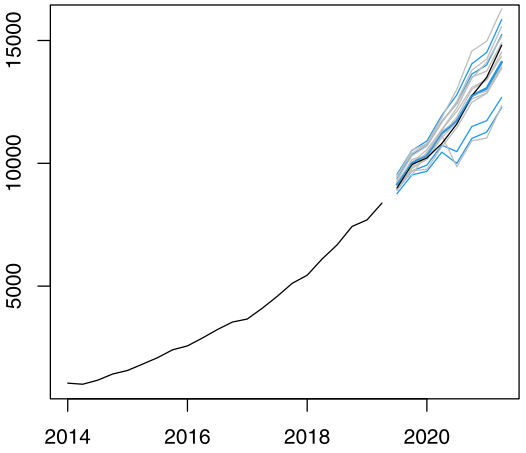


Fig. 3. Quarterly revenue of Amazon Web Services (in million USD, source: statista.com). LGT forecast two years ahead, ten examples of the forecasting trajectories (in light grey), and the 1st, 5th, 50th, 95th, and 99th percentiles (in blue). Actuals shown in black. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

8.4. Prediction intervals

The forecasts result from step-by-step simulation with, e.g., 5000 paths and aggregation by calculating quantiles. Fig. 3 shows a particular monthly time series, the first ten simulation paths, the 1st, 5th, 95th, and 99th percentiles, and the 50th percentile taken as the expected value.

The prediction interval coverage appears to be category-dependent. Table 6 shows coverage ranges for our proposed methods and analogous ranges of auto ETS, auto ARIMA, THETA, and MAPA with analytically calculated prediction intervals. Note that the original M3 participants are not compared here as they did not predict

Table 6
Results for probabilistic forecast evaluation. Below 99/95/5/1p – the percentage of cases when the true value turned out to be below the 99th/95th/5th/1st percentile; MSIS 90p/98p – MSIS for intervals of the size that contain 90/98 per cent of the observations, respectively.

	Below 99p	Below 95p	Below 5p	Below 1p	MSIS 90p	MSIS 98p
All (amount of series: 3003)						
{L,S}GT	**97.32**	**92.22**	6.37	2.22	**8.70**	**15.89**
ETS/ZZZ	95.87	90.75	**5.35**	**1.79**	9.79	20.10
ARIMA	96.18	91.49	8.07	3.76	11.31	27.43
THETA	95.37	89.87	6.28	2.50	10.09	22.02
MAPA	94.02	88.30	7.03	3.18	10.98	26.95
Yearly (amount of series: 645, horizon: 6)						
LGT	**97.16**	**91.42**	**6.23**	**2.04**	**17.38**	**32.64**
ETS/ZZZ	91.81	86.41	7.26	3.88	21.80	50.49
ARIMA	91.27	85.58	12.53	7.57	26.53	76.15
THETA	91.47	84.34	**6.23**	2.61	21.59	53.76
MAPA	87.42	81.27	9.69	6.23	25.78	74.64
Other (amount of series: 174, horizon: 8)						
LGT	99.43	97.49	4.60	**1.44**	**10.69**	**16.68**
ETS/ZZZ	98.20	**94.54**	6.11	2.16	10.93	18.02
ARIMA	**98.78**	95.47	5.46	2.95	12.37	19.99
THETA	99.57	97.56	9.55	3.16	13.32	20.64
MAPA	98.28	95.69	**5.32**	1.94	11.75	17.12
Monthly (amount of series: 1428, horizon: 18)						
SGT	**97.51**	**92.55**	**5.21**	1.69	**5.10**	**8.20**
ETS/ZZZ	96.55	91.37	4.28	**1.12**	5.17	8.42
ARIMA	97.04	92.37	5.93	2.31	5.54	9.94
THETA	95.81	90.35	5.69	2.26	5.62	10.25
MAPA	94.95	88.99	5.57	2.09	5.50	10.23
Quarterly (amount of series: 756, horizon: 8)						
SGT	**96.13**	90.16	11.79	4.76	**7.64**	**15.95**
ETS/ZZZ	95.02	90.01	8.50	**3.22**	7.99	16.71
ARIMA	95.11	**90.62**	14.90	7.64	8.97	20.61
THETA	95.04	89.60	**8.09**	3.29	7.98	17.47
MAPA	93.35	88.18	11.94	6.15	8.52	20.08

intervals, and baggedETS cannot produce prediction intervals. ETS and ARIMA produce too narrow intervals, which the table can confirm. While our method tends to produce too narrow intervals (for example, for yearly series forecasted, in 97.16% of cases, the forecasted values are lower than 99p, but this should have happened in 99% of cases), in the large majority of cases, the model performs better than the comparison methods. Interestingly, though not shown in this table, the simulated intervals of ETS and ARIMA are worse and even narrower. Checking the MSIS, which measures both calibration and sharpness of the intervals, we see that our proposed method consistently outperforms the comparison methods in all cases.

8.5. Ablation study and computation times

While the LGT and SGT approaches are based on ETS methods, they differ from them in multiple ways. The main differences are concentrated between ETS (AAN) and LGT, while SGT can be considered a direct extension of LGT into the space of seasonal data; the ETS model closest to SGT is the AAM model. Thus, in this ablation study, we investigate differences between ETS (AAN) and LGT for non-seasonal data to find the innovations' importance and whether the complete set of these innovations is required to achieve the reported performance. Seasonal data is studied similarly. We run performance testing on different variants of our proposed LGT and SGT approaches. The results of the ablation study are reported in Table 7. We remove one-by-one features such as Student t-distribution and heteroscedasticity for errors and replace them with constant variance and normal distribution, respectively. Also, we remove the global trend feature. We note that SGT without a global trend has no trend, as SGT has no local trend. As overall comparison methods, we report the results of a run of auto ETS (denoted ETS/ZZZ) and a run of auto ETS with a Box–Cox transformation as preprocessing. In the Box–Cox transformation, we choose the parameter λ with the method from (Guerrero, 1993), which is also implemented in the *forecast* package in R. As a deviation from the default behaviour of the implementation, we restrict λ to be non-negative, as otherwise we observe an explosive behaviour on some time series and the method returns *NA* values as forecasts. Furthermore, for non-seasonal data, we run our custom Gibbs sampler that is available for the non-seasonal case and further run ETS/AAN, ETS/AAN (Bayesian version) – ETS (AAN) fitted with Stan using our Bayesian approach, and an implementation of the LGT method within the *forecast* package in R, which is thus fitted using the standard optimisation method which is used to fit ETS methods. In terms of seasonal data, the *forecast* package in R does not allow the fitting of the ETS/AAM models due to numerical instabilities. We fit

Table 7
Performance on M3 data of the ablation study methods. Average runtime is given in seconds.

	sMAPE	MASE	MSIS 90p	MSIS 98p	Avg Runtime [s]
	All (amount of series: 3003)				
ETS/ZZZ (forecast R package)	13.07	1.43	9.79	20.10	0.30
ETS/{AAN,MAM} (forecast R package)	14.41	1.52	11.66	28.17	0.04
ETS/BoxCox (forecast R package)	13.28	1.53	10.81	23.86	0.01
{L,S}GT (our R package, Stan sampler)	**12.29**	**1.30**	**8.70**	**15.89**	368.03
{L,S}GT without global trend	13.04	1.45	9.34	16.07	271.47
{L,S}GT without heteroscedasticity	12.29	1.31	9.54	19.81	421.87
{L,S}GT without student distribution	12.41	1.31	8.74	16.11	298.19
	Yearly (amount of series: 645, horizon: 6)				
ETS/ZZZ (forecast R package)	17.00	2.86	21.80	50.49	0.01
ETS/AAN (forecast R package)	18.72	3.07	28.66	84.79	0.01
ETS/BoxCox (forecast R package)	18.00	3.24	25.88	66.81	0.00
LGT (our R package, Stan sampler)	15.18	**2.48**	17.38	32.64	50.40
LGT (our R package, Gibbs sampler)	**14.89**	2.54	18.46	46.97	3.79
LGT (using MLE within forecast R package)	21.49	3.29	27.77	73.49	0.01
LGT without global trend	16.23	2.71	17.52	**29.07**	18.95
LGT without heteroscedasticity	14.99	2.53	20.52	47.90	50.69
LGT without student distribution	15.35	2.50	**17.24**	32.36	37.51
ETS/AAN (Bayesian version)	15.94	2.81	21.76	48.85	14.33
	Other (amount of series: 174, horizon: 8)				
ETS/ZZZ (forecast R package)	4.37	1.81	10.93	18.02	0.01
ETS/AAN (forecast R package)	4.77	1.95	11.37	**15.79**	0.00
ETS/BoxCox (forecast R package)	4.35	1.89	11.28	17.36	0.00
LGT (our R package, Stan sampler)	4.25	1.72	**10.69**	16.68	161.51
LGT (our R package, Gibbs sampler)	**4.23**	**1.70**	11.28	21.43	5.19
LGT (using MLE within forecast R package)	5.01	2.13	15.82	22.86	0.02
LGT without global trend	4.60	2.03	12.23	18.23	55.38
LGT without heteroscedasticity	4.24	1.71	10.74	16.79	206.74
LGT without student distribution	4.34	1.79	11.46	17.58	90.83
ETS/AAN (Bayesian version)	4.71	2.09	12.96	19.97	39.48
	Monthly (amount of series: 1428, horizon: 18)				
ETS/ZZZ (forecast R package)	14.14	0.86	5.17	8.42	0.58
ETS/MAM (forecast R package)	15.54	0.92	5.64	9.37	0.08
ETS/BoxCox (forecast R package)	14.17	0.87	5.32	8.76	0.03
SGT (our R package, Stan sampler)	**13.77**	**0.83**	**5.10**	**8.20**	444.97
SGT without global trend	14.42	0.93	5.88	9.70	444.69
SGT without heteroscedasticity	13.87	**0.83**	5.29	8.89	561.22
SGT without student distribution	13.86	0.84	5.15	8.55	471.34
	Quarterly (amount of series: 756, horizon: 8)				
ETS/ZZZ (forecast R package)	9.68	1.17	7.99	16.71	0.08
ETS/MAM (forecast R package)	10.83	1.22	8.58	18.22	0.01
ETS/BoxCox (forecast R package)	9.63	1.22	8.21	17.23	0.01
SGT (our R package, Stan sampler)	**8.87**	**1.07**	**7.64**	**15.95**	541.21
SGT without global trend	9.64	1.22	8.22	16.50	209.46
SGT without heteroscedasticity	**8.87**	**1.07**	7.92	17.14	524.86
SGT without student distribution	9.02	1.08	7.65	16.18	241.28

MAM models instead, using multiplicative error instead of additive error, to compare with the ablation versions of SGT.

The results in Table 7 allow us to make the following observations. The model version without Student t-distribution is systematically worse than the full {L,S}GT model throughout all categories on both sMAPE and MASE. The differences are rather small for MASE. MSIS results also show similar results, such that the full version is better than the version without Student t-distribution overall and in almost all particular cases (except MSIS 90p for yearly and quarterly).

Regarding the model without heteroskedasticity, in terms of prediction intervals, which is the main reason for including heteroskedasticity, the model without heteroskedasticity is systematically worse in terms of MSIS than the full model. For point forecasting, the model is comparable, sometimes slightly better, and sometimes slightly worse. Removing the global trend from the models leads to systematically worse results by large margins. Comparing Bayesian ETS to its MLE counterparts, we observe that our implementation of Bayesian ETS/AAN outperforms both ETS/ZZZ and ETS/AAN on all measures in the yearly category but is generally worse than the MLE versions in the other category.

As a part of the ablation study, we also recorded average execution times (per single time series, rounded to 0.01 s) for the investigated methods. We can see that the Bayesian-based methods are slower than the original exponential smoothing techniques by orders of magnitude. But with the unimpressive performance of our implementation of LGT in the *forecast* package, the Bayesian model

fitting seems necessary to leverage the performance gains of the proposed models. Also note that the bespoke Gibbs sampler runs about an order of magnitude faster than the Stan implementation, with very competitive results, often outperforming the Stan implementation. We can further observe that removing the Student t-distribution and the global trend speeds up the models considerably. The model without Student t-distribution is still relatively performant; in particular, it can outperform ETS/ZZZ in all cases but monthly MSIS 98p. Thus, it can be a good choice for faster computation, though in the case of non-seasonal data, with full LGT and the bespoke Gibbs sampler, an even faster model is available.

9. Conclusions

We have presented two related models, the seasonal SGT and the non-seasonal LGT model, derived from the AAZ exponential smoothing models. They are designed to perform well on fast-growing and often irregular time series found in fast-growing businesses such as cloud services or rideshare businesses and on the M3 competition dataset. Here, they bested the top-performing algorithms in every category/metric pair and auto ARIMA and auto ETS from the *forecast* R package. We believe that the very good forecasting performance, in terms of point- and probabilistic forecasting of the models on the M3 dataset, comes mostly from their use of flexible, nonlinear functions for the trend and the size of the error. Standard Exponential Smoothing models can only be either linear or exponential in their trend – these models can deliver both extremes and anything in between. Similarly, the function for the size of the error bridges Exponential Smoothing models with additive and multiplicative errors. A Student's t-distribution for the error term is also more capable of fitting to both smooth and fat-tailed time series. Probabilistic programming systems, including Stan, provide a convenient platform for the development of innovative time series algorithms.

The main drawback of our new methods is that the Stan-based methods are orders of magnitude (typically over a thousand times) slower than the original ETS techniques. Though we have overcome this problem to a certain extent by implementing a bespoke Gibbs sampler, shaving off one order of magnitude for the non-seasonal case, our methods are still at least hundreds of times slower than the classical ETS methods. However, in the new forecasting world where large-scale forecasting is typically performed very competitively by global cross-series models (often Machine Learning models), our method can have an important place in the forecaster's toolbox to tackle forecasting a small number of series with low granularity and without covariates, in a generally data-scarce environment.

CRediT authorship contribution statement

Slawek Smyl: Conceptualization, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Christoph Bergmeir:** Conceptualization, Investigation, Methodology, Project administration, Software, Supervision, Writing – original draft, Writing – review & editing. **Alexander Dokumentov:** Methodology, Software, Validation, Writing – original draft. **Xueying Long:** Investigation, Software, Validation, Writing – review & editing. **Erwin Wibowo:** Methodology, Software, Writing – original draft. **Daniel Schmidt:** Conceptualization, Formal analysis, Methodology, Software, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Christoph Bergmeir reports financial support was provided by University of Granada, Ministry of Universities (Spain), Next Generation Funds from the European Union, and from the Australian Research Council. Alexander Documentov and Erwin Wibowo report financial support was provided by Monash University. Christoph Bergmeir is an associate editor of the International Journal of Forecasting.

Acknowledgments

Christoph Bergmeir is supported by a María Zambrano (Senior) Fellowship that is funded by the Spanish Ministry of Universities and Next Generation funds from the European Union. He was also supported by the Australian Research Council under grant DE190100045 for part of the work. Erwin Wibowo undertook the work on the paper while affiliated with Monash University. Alexander Dokumentov was funded by Monash University, Australia to undertake this work.

References

Andrawis, R., & Atiya, A. (2009). A new Bayesian formulation for Holt's exponential smoothing. *Journal of Forecasting*, *28*(3), 218–234.

Beaumont, A. N. (2014). Data transforms with exponential smoothing methods of forecasting. *International Journal of Forecasting*, *30*(4), 918–927.

Benavoli, A., Corani, G., & Mangili, F. (2016). Should we really use post-hoc tests based on mean-ranks?. *Journal of Machine Learning Research*, *17*(1), 152–161.

Bergmeir, C., Hyndman, R. J., & Benítez, J. M. (2016). Bagging exponential smoothing methods using stl decomposition and Box-Cox transformation. *International Journal of Forecasting*, *32*, 303–312.

Bermudez, J., Corberán-Vallet, A., & Vercher, E. (2009). Multivariate exponential smoothing: A Bayesian forecast approach based on simulation. *Mathematics and Computers in Simulation*, *79*(5), 1761–1769.

Bermudez, J., Segura, J., & Vercher, E. (2010). Bayesian forecasting with the Holt-Winters model. *Journal of the Operational Research Society*, *61*(1), 164–171.

Betancourt, M. (2017). A conceptual introduction to Hamiltonian Monte Carlo. arXiv preprint arXiv:1701.02434.

Box, G. E. P., & Cox, D. R. (1964). An analysis of transformations. *Journal of the Royal Statistical Society. Series B*, *26*(2), 211–252.

Boylan, J., Babai, M., Mohammadipour, M., & Syntetos, A. (2015). Reproduction of simple forecasting methods in the M competitions. In *35th international symposium on forecasting*. Riverside (California).

Bunse, M. (2023). critdd | Critical difference diagrams. URL: https://github.com/mirkobunse/critdd.

Cipra, T. (1992). Robust exponential smoothing. *Journal of Forecasting*, *11*(1), 57–69.

Corberán-Vallet, A., Bermúdez, J. D., & Vercher, E. (2011). Forecasting correlated time series with exponential smoothing models. *International Journal of Forecasting*, *27*(2), 252–265.

Corberan-Vallet, A., Bermudez, J., & Vercher, E. (2013). Bayesian forecasting of demand time-series data with zero values. *European Journal of Industrial Engineering*, *7*(6), 777–796.

Crevits, R., Bergmeir, C., Hyndman, R., Ihaka, R., & R Core Team (2018). robets: Forecasting time series with robust exponential smoothing. R package version 1.4. URL: https://cran.r-project.org/package=robets.

Crevits, R., & Croux, C. (2016). Forecasting with robust exponential smoothing with damped trend and seasonal components.

Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *The Journal of Machine learning research*, *7*(1), 1–30.

Fahrmeir, L., & Kaufmann, H. (1985). Consistency and asymptotic normality of the maximum likelihood estimator in generalized linear models. *The Annals of Statistics*, *13*(1), 342–368.

Gardner, E. (1985). Exponential smoothing: The state of the art. *Journal of Forecasting*, *4*(1), 1–28.

Gardner, E., Jr. (2006). Exponential smoothing: The state of the art-part II. *International Journal of Forecasting*, *22*(4), 637–666.

Gelman, A. (2006). Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper). *Bayesian Analysis*, *1*(3), 515–534.

Gelper, S., Fried, R., & Croux, C. (2010). Robust forecasting with exponential and Holt-Winters smoothing. *Journal of Forecasting*, *29*(3), 285–300.

Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American statistical Association*, *102*(477), 359–378.

Goodwin, P. (2010). The Holt-Winters approach to exponential smoothing: 50 years old and going strong. *Foresight: The International Journal of Applied Forecasting*, *19*, 30–33.

Guerrero, V. (1993). Time-series analysis supported by power transformations. *Journal of Forecasting*, *12*, 37–48.

Hoffman, M., & Gelman, A. (2014). The no-u-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, *15*, 1593–1623.

Hyndman, R. J., Akram, M., & Archibald, B. C. (2008). The admissible parameter space for exponential smoothing models. *Annals of the Institute of Statistical Mathematics*, *60*, 407–426.

Hyndman, R., Akram, M., & Bergmeir, C. (2013). Mcomp: Data from the M-competitions. URL: http://robjhyndman.com/software/mcomp/.

Hyndman, R., & Athanasopoulos, G. (2018). *Forecasting: principles and practice* (2nd ed.). Melbourne, Australia: OTexts, https://OTexts.com/fpp2. (Accessed on 30 April 2019).

Hyndman, R., & Khandakar, Y. (2008). Automatic time series forecasting: The forecast package for R. *Journal of Statistical Software*, *27*(3), 1–22.

Hyndman, R., & Koehler, A. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, *22*(4), 679–688.

Hyndman, R. J., Koehler, A. B., Ord, J. K., & Snyder, R. D. (2008). *Springer series in statistics*, *Forecasting with exponential smoothing: The state space approach*. Springer, URL: http://www.exponentialsmoothing.net.

Hyndman, R., Koehler, A., Snyder, R., & Grose, S. (2002). A state space framework for automatic forecasting using exponential smoothing methods. *International Journal of Forecasting*, *18*(3), 439–454.

Januschowski, T., Gasthaus, J., Wang, Y., Salinas, D., Flunkert, V., Bohlke-Schneider, M., et al. (2020). Criteria for classifying forecasting methods. *International Journal of Forecasting*, *36*(1), 167–177.

Kourentzes, N., Petropoulos, F., & Trapero, J. (2014). Improving forecasting by estimating time series structural components across multiple frequencies. *International Journal of Forecasting*, *30*(2), 291–302.

Lange, K. L., Little, R. J. A., & Taylor, J. M. G. (1989). Robust statistical modeling using the *t* distribution. *Journal of the American Statistical Association*, *84*(408), 881–896.

Lunn, D. J., Thomas, A., Best, N., & Spiegelhalter, D. (2000). WinBUGS & ndash; a Bayesian modelling framework: Concepts, structure, and extensibility. *Statistics and Computing*, *10*(4), 325–337.

Makridakis, S., & Hibon, M. (2000). The M3-competition: Results, conclusions and implications. *International Journal of Forecasting*, *16*(4), 451–476.

Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2018). The M4 competition: Results, findings, conclusion and way forward. *International Journal of Forecasting*, *34*(4), 802–808.

Neal, R. (1994). An improved acceptance procedure for the hybrid Monte Carlo algorithm. *Journal of Computational Physics*, *111*(1), 194–203.

O'Hara-Wild, M., Hyndman, R., Wang, E., Caceres, G., Hensel, T.-G., & Hyndman, T. (2021). fable: Forecasting models for tidy time series. R package version 0.3.1. URL: https://cran.r-project.org/package=fable.

Parzen, E. (1982). Ararma models for time series analysis and forecasting. *Journal of Forecasting*, *1*(1), 67–82.

Plummer, M. (2003). JAGS: A program for analysis of Bayesian graphical models using Gibbs sampling. In *3rd international workshop on distributed statistical computing, vol. 124*.

Pritularga, K. F., Svetunkov, I., & Kourentzes, N. (2023). Shrinkage estimator for exponential smoothing models. *International Journal of Forecasting*, *39*(3), 1351–1365.

R-bloggers (2014). JAGS and Stan. https://www.r-bloggers.com/jags-and-stan/. (Accessed 30 April 2019).

R Core Team (2018). *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing, URL: https://www.R-project.org/.

Sbrana, G., & Silvestrini, A. (2020). Forecasting with the damped trend model using the structural approach. *International Journal of Production Economics*, *226*, Article 107654.

Schmidt, D., & Makalic, E. (2019). Bayesian generalized horseshoe estimation of generalized linear models. In *Machine learning and knowledge discovery in databases: European conference*.

Stan Development Team (2015). Stan modeling language users guide and reference manual, version 2.9.0. URL: http://mc-stan.org/.

Stan Development Team (2018). RStan: The R interface to Stan. R package version 2.18.2. URL: http://mc-stan.org/.

Svetunkov, I., Kourentzes, N., & Ord, J. K. (2022). Complex exponential smoothing. *Naval Research Logistics*, *69*(8), 1108–1123.

Tsionas, M. G. (2021). Bayesian forecasting with the structural damped trend model. *International Journal of Production Economics*, *234*, Article 108046.

Wand, M. P., Ormerod, J. T., Padoan, S. A., & Fruhwirth, R. (2011). Mean field variational Bayes for elaborate distributions. *Bayesian Analysis*, *6*(4), 847–900.